

*“The limits of computable functions lie not in nature,
but in our definitions.”*

— Alan Turing, 1936

The Arithmetic of Order: A Computable Foundation for Physics

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Dedication

*This work is dedicated to **Alan Mathison Turing** (1912–1954),
whose pioneering insights into the limits of logic, the nature of computation,
and the generative power of discrete processes laid the foundation for
a new understanding of mathematics, physics, and intelligence.*

*The Arithmetic of Order (AoO) seeks to extend Turing’s vision:
that finite operations, governed by internal logic,
can unfold into the full richness of the physical world.*

Abstract

This paper presents the Arithmetic of Order (AoO), a framework for foundational physics built on the single principle that reality emerges from a discrete, computable, and non-associative algebraic substrate over the rational numbers. We formalize an *Origin–Observation Duality*, where a single tri-octonion state vector v generates a unique Hermitian Jordan operator A that satisfies both a nonlinear internal eigen-equation ($Av = v[v]$) and admits a classical rational-valued spectral decomposition ($Aw = \lambda w, \lambda \in \mathbb{Q}$). The dual nature of this spectrum provides a deductive origin for an emergent 3+1D spacetime with an intrinsic particle/anti-particle symmetry. We apply this framework to reformulate LHC jet production as the spectral footprint of a G_2 automorphism cascade, a non-perturbative process governed by octonion algebra. Finally, we show how the core principles of AoO are mirrored in the architectures of modern generative models, grounding the theory in experimental and computational practice.

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Introduction

This paper introduces the Arithmetic of Order (AoO), a framework for foundational physics built from a single, computable principle [5, 1]. We reject the continuum as axiomatic and instead propose that physical reality emerges from a discrete, non-associative algebraic structure over the rational numbers.

The core claims of this work are as follows:

1. We formalize an **Origin–Observation Duality**, where a single algebraic state vector v deterministically generates a unique physical operator A through a nonlinear internal logic ($Av = v[v]$), while the same operator produces quantized, measurable observables through a standard linear spectral decomposition ($Aw = \lambda w$).
2. We demonstrate that the spectral properties of the operator A give rise to an **emergent 3+1D spacetime with an intrinsic particle/anti-particle duality**, without requiring additional physical postulates.
3. We apply this framework to provide a deterministic, non-perturbative model for **hadron jet production** at the LHC, reformulating it as the spectral signature of a discrete cascade of G_2 automorphisms.

This work aims to build a bridge between the foundations of computation, algebra, and physics, proposing a universe that is, in its essence, a computation.

1 Background and Historical Context

The Arithmetic of Order (AoO) draws deeply on the legacy of foundational thinkers—from Hamilton and Cayley, pioneers of hypercomplex numbers, through Pascual Jordan, who alongside John von Neumann and Eugene Wigner developed Jordan algebras instrumental in quantum mechanics [11], to Alan Turing, whose groundbreaking insights into computability laid the groundwork for modern computational theory. John von Neumann’s work on the mathematical foundations of quantum mechanics [12] and operator algebras provides a critical undercurrent in the algebraic structure AoO utilizes. Equally foundational are the developments in set theory and formal axiomatic systems, driven by Ernst Zermelo [13] and Abraham Fraenkel [14], who formulated the ZF axioms, and David Hilbert, whose program sought rigorous axiomatization of all mathematics [15]. Moreover, Kurt Gödel’s incompleteness theorems [16] cast a lasting philosophical shadow by demonstrating intrinsic limitations in formal systems, underscoring the importance of the computable and finitistic approaches that AoO embraces. By situating AoO within this rich tapestry of mathematical and logical advancements, the framework honors and extends the profound work that shaped the modern understanding of algebra, computation, and physical law.

2 The Core Algebraic Engine: A Finitistic Approach

2.1 The Tri-Octonion State and the Associator over $\mathbb{Q}$

The fundamental unit of information in AoO is a **tri-octonion state vector** over the rational numbers, an ordered triple of pure imaginary rational octonions [2]:

$$v = (o_1, o_2, o_3) \in \text{Im}(\mathbb{O}_{\mathbb{Q}})^3$$

Its intrinsic topological structure is quantified by the **associator**:

$$[v] := (o_1 o_2) o_3 - o_1 (o_2 o_3) \in \text{Im}(\mathbb{O}_{\mathbb{Q}})$$

This non-zero "non-associative charge" is the engine of all dynamics. The framework is "finite yet unbounded" as it is built upon $\mathbb{Q}$, where every element is finitely representable, yet the set itself has no upper bound, reflecting the generative principle $1 \rightarrow n \rightarrow n + 1$.

2.2 The Origin–Observation Duality

The central mechanism of AoO is a duality that separates the internal, generative logic of a system from its external, measurable manifestation.

Proposition 2.1 (Origin–Observation Duality). *Let $v = (o_1, o_2, o_3) \in \text{Im}(\mathbb{O}_{\mathbb{Q}})^3$ be a tri-octonion state with associator $[v]$. Let $A \in \text{Herm}_3(\mathbb{O}_{\mathbb{Q}})$ be the unique Hermitian Jordan operator constructed from v and six rational parameters. This operator has the form:*

$$A = \begin{pmatrix} p & a & \bar{c} \\ \bar{a} & m & b \\ c & \bar{b} & n \end{pmatrix}$$

with $p, m, n \in \mathbb{Q}$ and $a, b, c \in \mathbb{O}_{\mathbb{Q}}$. The following duality holds:

1. **Origin of Law (Internal Logic):** The operator A is selected by the nonlinear, self-referential eigen-equation:

$$Av = v[v]$$

This equation defines the internal, unobservable law consistent with the state's algebraic topology. The associator $[v]$, an octonion, acts as a hidden structural eigenvalue, making the law uniquely determined by its own state.

2. **Observation of Law (External Measurement):** The same operator A , as an element of a formally real Jordan algebra over $\mathbb{Q}$, has a classical spectral decomposition yielding quantized observables [4]:

$$Aw = \lambda w, \quad \lambda \in \mathbb{Q}, \quad w \in \mathbb{O}_{\mathbb{Q}}^3,$$

where the eigenvalues $\{\lambda_i\}$ are rational scalars and the eigenvectors $\{w^{(i)}\}$ are orthogonal. These correspond to the quantized, measurable observables of the system.

Remark 2.2 (Foundational Acknowledgement). The construction of the operator A is an adaptation over $\mathbb{Q}$ of the foundational work by Gillow-Wiles and Dray [6], who proved the existence of such a matrix family over $\mathbb{R}$. In AoO, their mathematical result is reinterpreted as a physical law-generation mechanism within a computable, finitistic framework.

3 Emergent Structure: Spacetime and Particle Duality

In the standard, continuum-based account—most familiarly via the Dirac equation—anti-particles appear as a somewhat ad hoc solution to the problem of negative-energy states. One posits their existence to make the formalism work, but the origin of that duality remains mysterious.

By contrast, the Arithmetic of Order (AoO) delivers particle–anti-particle symmetry as a direct consequence of its discrete, non-associative algebra. A single tri-octonion state carries within it two complete 3-dimensional ‘frames’—each defined by one of the two triples of real (here, rational) eigenvalues of the associated Hermitian Jordan matrix A . One eigen-triple yields the particle frame, the other yields its anti-particle counterpart.

Thus anti-particles are no longer an external addendum but an intrinsic feature of the algebraic substrate. Charge conjugation, for instance, becomes nothing more mystical than the switch from one eigen-frame to the other. In AoO, particle and anti-particle are indeed “two sides of the same coin,” emerging naturally from the granular structure of reality rather than grafted on to save the theory.

Corollary 3.1 (Intrinsic Spectral Duality). *Let $A \in \text{Herm}_3(\mathbb{O}_{\mathbb{Q}})$ be the Jordan operator generated by a valid tri-octonion seed. The spectral decomposition of A via the standard eigenvalue problem is intrinsically dual: it necessarily yields two complete, Jordan-orthogonal 3-frames. These frames are not independent but are inseparable co-products of the single operator A , representing the mingled particle and anti-particle aspects of the system.*

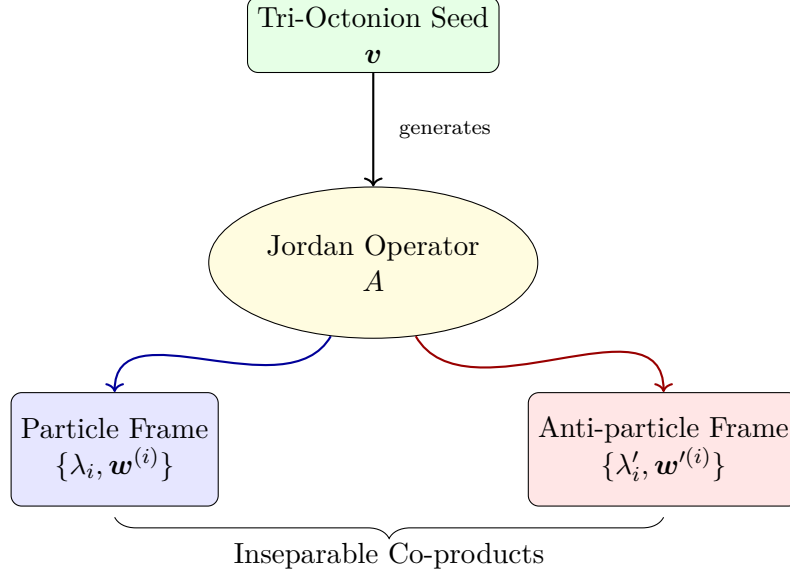


Figure 1: The Intrinsic Duality of the AoO Spectrum. A single operator A , generated from a seed v , necessarily unfolds into two mingled spectral frames, corresponding to a particle/anti-particle system.

3.1 The Dual-Frame Eigenvalue Structure

A key result for 3×3 Hermitian octonionic matrices is that the standard eigenvalue problem $Aw = \lambda w$ yields a spectrum of six real (or, in AoO, rational) eigenvalues. These eigenvalues naturally partition into two distinct families of three, defining a dual-frame structure. This is not an assumption of the model but a direct consequence of the non-associative algebra.

- **Dual 3D Frames (Particle/Anti-particle Space):** The six eigenvalues $(\lambda_1, \lambda_2, \lambda_3)$ and $(\lambda'_1, \lambda'_2, \lambda'_3)$, along with their corresponding orthonormal eigenvectors, define two complete and distinct 3D reference frames. As formalized in the corollary above, this algebraic duality is interpreted as the physical duality between a ****particle**** and its ****anti-particle****.
- **1D Time (The Algebraic Clock):** The evolution of the system, $v_n \rightarrow v_{n+1}$, is driven by the irreversible dynamics of the state's internal logic. The sequence of associators, $[v_n]$, serves as a discrete and intrinsic algebraic clock.

Thus, a 3+1 dimensional structure with an inherent particle/anti-particle symmetry emerges directly from the AoO framework.

3.2 The Turing Emergence Theorem

This entire process, where a discrete seed generates a rich, dualistic, and observable reality, is formalized by the Turing Emergence Theorem.

Theorem 3.2 (Turing Emergence). *Let a physical system be defined by a finite tri-octonion seed $v \in \text{Im}(\mathbb{O}_{\mathbb{Q}})^3$ with a non-zero associator. The Arithmetic of Order (AoO) dictates that this seed deterministically generates a unique Hermitian Jordan operator $A \in \text{Herm}_3(\mathbb{O}_{\mathbb{Q}})$. The classical spectral decomposition of A yields a set of six rational, measurable eigenvalues that define a dual 3D reference frame, corresponding to a particle/anti-particle system.*

Thus, a discrete, internal, and computable logic deterministically unfolds into an observable, externally consistent physical reality with inherent dualities. This realizes Turing's vision of universal computation manifesting as physical law.

Remark 3.3 (Analogy to Turing's Oracle). This process mirrors Turing's own model of computation. The internal, computable rules governing the AoO state act as a "Turing Machine." The spectral decomposition, which provides answers (the eigenvalues) that are not contained within the initial rules, functions as an "Oracle." The theorem thus frames physical law as an interplay between a system's internal computation and its observable, oracle-like output.

4 Reformulating Hadron Collider Physics

4.1 Algebraic Jet Emission via G_2 Automorphism Cascade

AoO provides a deterministic, non-perturbative model for jet production at the LHC. A jet is the spectral footprint of a discrete algebraic transformation.

1. **Initial State:** A quark is modeled as a tri-octonion state v .
2. **Interaction as Automorphism:** A gluon-like interaction is modeled by the action of a G_2 automorphism, $F(S, T, X)$, which transforms the state $v \rightarrow v'$.
3. **Jet Cascade:** Hadronization is a deterministic cascade of these automorphisms, $v \rightarrow v' \rightarrow v'' \dots$.
4. **Observable Jet:** The jet is the collection of the spectral outputs from the sequence of difference operators, $\Delta A_k = A^{(k)} - A^{(k-1)}$.

This model makes concrete physical predictions. The energy of the emitted jet components corresponds directly to the rational eigenvalues $\{\lambda_i\}$ of the difference operators ΔA_k . Furthermore, the angular distribution of the jet is determined by the geometry of the corresponding eigenvectors $\{w^{(i)}\}$, which are fixed by the sequence of G_2 automorphisms. The framework thus predicts a discrete energy spectrum and quantized angular correlations for hadron jets.

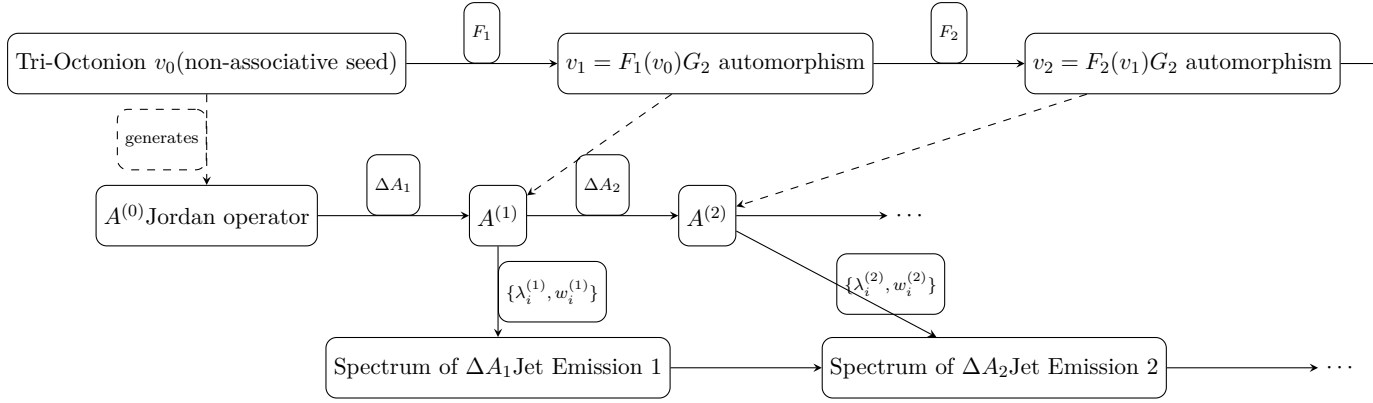


Figure 2: Algebraic jet cascade in the Arithmetic of Order: tri-octonion states evolve via G_2 automorphisms, generating Jordan operators $A^{(k)}$, whose spectral differences ΔA_k correspond to emitted jets with measurable energy and direction spectra.

5 A Foundational Critique of Modern Computational Models

The success of modern AI models in particle physics, such as autoregressive transformers for jet simulation [9] and neural unfolding procedures [10], is undeniable. These tools exhibit impressive power in data generation and analysis. However, from the perspective of the Arithmetic of Order, they are sophisticated descriptive models built upon the existing perturbative and continuum-based paradigm, rather than being truly deductive, generative foundations. Their limitations highlight the need for a deeper, non-perturbative algebraic structure.

5.1 Autoregressive Models: Correct Intuition, Ad Hoc Rules

The architecture of autoregressive models like JetGPT [9] correctly intuit a core principle of AoO: that physical processes like jet hadronization are fundamentally **generative and sequential**. They build a jet piece-by-piece, with each new particle emission conditioned on the previous ones, mirroring the AoO principle of a state evolving from $1 \rightarrow n \rightarrow n + 1$.

However, the weakness of this approach lies in its origin. The rules governing the generation are not derived from a fundamental algebraic principle but are **learned by mimicking data** from existing simulations (e.g., Pythia, Herwig). These simulations are themselves rooted in **perturbative QCD** matched with phenomenological hadronization models. Therefore, while the *structure* of the AI model aligns with AoO's generative vision, its *logic* is ad hoc and inherits the limitations and approximations of the perturbative framework it is trained on. It is a powerful mimic, not an *ab initio* theory.

5.2 Unfolding Procedures and the Flaw of the Continuum

Neural unfolding techniques like SPINUP [10] address the critical problem of correcting for detector effects, which "blur" or "smear" the true physical data. This approach, while innovative, is even more fundamentally ad hoc from the AoO perspective.

The very concept of "un-blurring" a signal presupposes that the underlying reality is a **continuous signal that has been distorted**. It employs sophisticated mathematical tools rooted in continuum analysis to solve this problem. In contrast, AoO posits that physical reality is **fundamentally discrete and quantized**. The true output of an interaction is not a continuous function but a finite set of rational eigenvalues $\{\lambda_i\}$. In this view, the measurement problem is not about "un-blurring" a continuum, but about correctly registering discrete events. 'SPINUP' is a brilliant tool for correcting a flawed, continuum-based model of measurement, not a reflection of a truly discrete reality.

5.3 The Need for a Non-Perturbative Foundation

These state-of-the-art computational methods reveal the limits of the current paradigm. They are powerful tools for working within it, but they cannot derive its structure from first principles.

The Arithmetic of Order provides the missing foundation. It replaces the ad hoc, learned rules of AI models with a deterministic, **non-perturbative algebraic engine**: the G_2 automorphism cascade and the Origin-Observation Duality. This provides a true generative model where the sequence of physical states and their measurable outputs are derived directly from the internal logic of octonion algebra. Such a foundation is not only more fundamental but could ultimately serve as the basis for a new generation of AI models designed not merely to mimic data, but to compute physical reality from its algebraic core.

6 From Philosophy to Prediction: A Synthesis

The Arithmetic of Order (AoO) is more than a descriptive model—it is a generative, first-principles framework for physics. Its credibility rests on the coherence of a single thread: from a finite algebraic substrate to testable predictions of physical reality. This section synthesizes that logical bridge.

The journey begins with a conceptual shift: AoO discards the continuum and adopts a **finitistic, non-associative algebra** over the rational numbers. Within this framework, the deep structures of nature are not postulated but arise by internal necessity. Nowhere is this more striking than in the case of **particle–anti-particle duality**, which emerges directly from the dual spectral frames of the theory’s core Jordan operators.

The algebraic substrate is **computable by design**. The nonlinear internal law $Av = v[v]$ functions as a self-contained “Turing machine,” governed by the state’s intrinsic topology. The linear spectral equation $Aw = \lambda w$ serves as an **oracle**, exposing rational eigenvalues not explicitly encoded in the state, but deterministically implied by it. This internal–external duality reframes the measurement problem: to observe is to query the oracle of a system’s internal logic.

From this principle, rich physical consequences unfold. The same associator that drives the state’s evolution also becomes the **source of algebraic torsion**, the generator of dynamics. The same spectral architecture that yields spacetime and duality also underpins a **non-perturbative model of hadron jets**, predicting rational energy spectra as eigenvalue differences of evolving operators.

In sum, AoO offers what modern physics has sought: a single, computable principle that generates spacetime, particle structure, and interaction dynamics—without invoking arbitrary constants, symmetry postulates, or continuum artifacts. The continuum is not fundamental—it is an illusion that dissolves under algebraic light.

7 Conclusion and Outlook

The Arithmetic of Order offers a complete, computable, and philosophically coherent foundation for physics. By reinterpreting the algebraic structures of the octonions and exceptional Jordan algebras over the rational numbers, it provides a deterministic origin for spacetime, particle states, and their interactions. The framework makes concrete, falsifiable predictions for hadron collider physics and finds deep resonance with the architectures of modern generative AI. Ultimately, AoO extends the legacy of Alan Turing, proposing a universe that is not merely described by computation but is, in its essence, a computation.

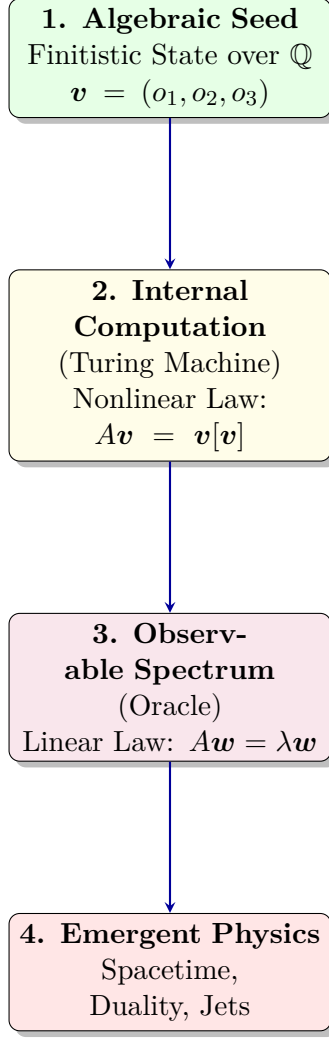


Figure 3: The AoO Pipeline: From a single algebraic seed, a deterministic computation generates an operator whose spectral properties (the oracle) unfold into the full structure of observable physics.

A Appendix: Oracle Automorphisms and Klein Character Varieties

The dynamics of the AoO lattice are regulated by higher-order algebraic structures that can be interpreted as "oracles."

A.1 The Automorphism $F(S, T, X)$ as an Interaction Operator

The map $F(S, T, X) := (TS)^{-1}(T(SX))$ is a G_2 automorphism that acts as a gluon-like operator, transforming the color state of a tri-octonion. This is based on the description of inner automorphisms of the octonions discussed by Baez and others [8].

A.2 Generation of Braid Group Symmetries

Compositions of these order-2 automorphisms generate order-3 symmetries of the form PXP^{-1} , providing an algebraic origin for the braid group B_3 and, consequently, the $SU(3)$ color symmetry of QCD [3].

A.3 The Klein Cubic Surface as a Manifold of Observables

The space of consistent interaction configurations is constrained by a character variety, the **Klein cubic surface**: $xyz + x^2 + y^2 + z^2 = x + y + z$. The discrete, 7-point braid group orbits on this surface correspond to distinct, quantized families of jet topologies.

B Appendix: Impact Analysis of the Turing Emergence Theorem

When: A Roadmap for Adoption and Validation

The validation of a new foundational framework typically unfolds through sequential phases. However, with the accelerating power of ensemble-based artificial intelligence—especially systems capable of symbolic reasoning and algebraic inference—the Arithmetic of Order (AoO) may progress through these phases far more rapidly than traditional paradigms.

- **Foundational Development and Simulation.** Theoretical refinement of the AoO framework continues in tandem with the development of open-source, algebraically grounded simulation tools. These tools allow researchers and AI agents to explore the dynamics of tri-octonion states, associators, and spectral unfolding.
- **Experimental and Computational Integration.** Collaborative efforts emerge between theorists, experimental physicists, and AI researchers to test AoO's falsifiable predictions. These include distinct spectral bandings, forbidden zones in collider data, and topological jet classifications. Simultaneously, ensemble AIs trained on AoO principles begin to serve as "inductive companions," guiding data analysis and unfolding processes with algebraic awareness.
- **Paradigm Integration and Technological Application.** As validation accumulates and simulation fidelity increases, AoO may become a unifying language for physics education, computation, and technology. Its non-associative logic could inspire novel quantum architectures, and its computable grammar might seed the next generation of AI models designed not merely to predict data, but to discover laws.

Rather than fixed timelines, the success of this roadmap hinges on the synergy between computable theory and intelligent agents—where both evolve together. AoO not only benefits from this AI acceleration but also informs its design, potentially making the framework concretely sound in the near future.

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